

ECE 107: Numerical Capacitance Extraction

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1 Introduction

Given two conductors with the potential on each surface given by $V_{top} = \frac{V_0}{2}$ and $V_{bottom} = -\frac{V_0}{2}$, we can formulate the integral equation for the unknown surface charge density distribution on the plates as

$$\iint_{S^{(1)}+S^{(2)}} \frac{1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|} \rho_s(\mathbf{r}') dS' = \begin{cases} V_0/2 & \mathbf{r} \in S_1 \\ -V_0/2 & \mathbf{r} \in S_2 \end{cases}$$

From this integral equation, we can discretize the problem by dividing the surfaces into $N = N_{S^{(1)}} + N_{S^{(2)}}$ patches, where $N_{S^{(1)}}, N_{S^{(2)}}$ are the numbers of patches on surfaces $S^{(1)}, S^{(2)}$. We then obtain the matrix equation

$$\sum_{n=1}^N \iint_{S_n} \frac{\rho_s(\mathbf{r}') dS'}{4\pi\epsilon_0 |\mathbf{r}_m - \mathbf{r}'|} = \begin{cases} V_0/2 & \mathbf{r}_m \in S_1 \\ -V_0/2 & \mathbf{r}_m \in S_2 \end{cases} \Rightarrow \sum_{n=1}^N Z_{mn} Q_n = V_m \Rightarrow \mathbf{Z}\mathbf{Q} = \mathbf{V}$$

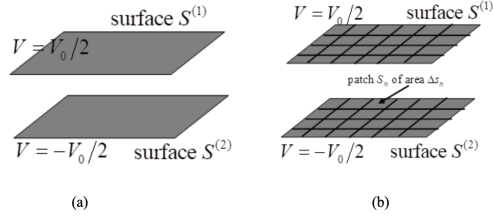


Figure 1: (a) Parallel plate capacitor; (b) Discretized problem.

Using these equations, we can find the discretized charges Q_n and capacitance C .

$$\mathbf{Q} = \mathbf{Z}^{-1}\mathbf{V} \text{ where } Q_n = \rho_s(\mathbf{r}_n)\Delta s_n.$$

$$C = \frac{\sum_{n=1}^{N_{S_1}} Q_n}{V_0}$$

2 Finding Charge and Capacitance

Using the above equations, we can use MATLAB to find the discretized charges Q_n and capacitance C . The MATLAB code used to find these values are as follows:

```
clear; clc;

epsilon = 8.85e-12;

V0 = 1;
d = 0.1;
w = 1;

nx = 10;
ny = 10;

M = nx*ny;
N = 2*M;

dx = w/nx;
dy = w/ny;

ds = dx*dy;

r = zeros(N,3);

cnt = 0;

for k = 1:2

    for n = 1:nx

        for m = 1:ny

            cnt = cnt + 1;

            r(cnt,1) = n*dx;
            r(cnt,2) = m*dy;

            if k == 1
                r(cnt,3) = d/2;
            else
                r(cnt,3) = -d/2;
            end

        end

    end

end

Zmm = zeros(N,N);

for m = 1:N

    for n = 1:N
```

```

    if m == n

        Zmn(m,n) = 1/(2*epsilon*sqrt(pi*ds));

    else

        R = norm(r(m,:) - r(n,:));

        Zmn(m,n) = 1/(4*pi*epsilon*R);

    end

end

end

V = zeros(N,1);

V(1:M) = V0/2;
V(M+1:N) = -V0/2;

Q = Zmn\V;

C = sum(Q(1:M))/V0;

fprintf( '\nCapacitance:\n' );
fprintf( 'C= %0.6e F\n\n', C);

fprintf( 'First 10 discretized charges:\n' );
fprintf( 'Patch-----Q_n [C]\n' );
fprintf( '-----\n' );

for i = 1:10
    fprintf( '%3d-----%0.6e\n', i, Q(i));
end

```

Running the MATLAB code gives us the following output for capacitance and charge:

```
Capacitance:
C = 1.170063e-10 F

First 10 discretized charges:
Patch      Q_n [C]
-----
1          1.613199e-12
2          1.367637e-12
3          1.347140e-12
4          1.337408e-12
5          1.333768e-12
6          1.333768e-12
7          1.337408e-12
8          1.347140e-12
9          1.367637e-12
10         1.613199e-12
```

Figure 2: MATLAB Command Window output for finding discretized charges and capacitance

We can see the list of multiple discretized charges, as well as the capacitance of our parallel plate capacitor to be about 0.12 nF.

3 Representation of Surface Charge Distribution

Using the parameters of $w = 1$ cm and $d = 3$ mm, we can generate a 3D representation of the surface charge distribution on one of the plates using the "surf" function in MATLAB. In this case, I chose the top plate. I also used a patch size of 10 to ensure that the size is not greater than d .

The MATLAB code to generate the figure is:

```
clear; clc; close all;

epsilon = 8.85e-12;

V0 = 1;

w = 1e-2;           % 1 cm
d = 3e-3;           % 3 mm

nx = 10;
ny = 10;

dx = w/nx;
dy = w/ny;
ds = dx*dy;

M = nx*ny;
N = 2*M;

r = zeros(N,3);

cnt = 0;

for k = 1:2
    for iy = 1:ny
        for ix = 1:nx
            cnt = cnt + 1;

            r(cnt,1) = -w/2 + (ix - 0.5)*dx;
            r(cnt,2) = -w/2 + (iy - 0.5)*dy;

            if k == 1
                r(cnt,3) = d/2;
            else
                r(cnt,3) = -d/2;
            end
        end
    end
end

Zmm = zeros(N,N);

for m = 1:N
    for n = 1:N

        if m == n
```

```

        Zmn(m,n) = 1/(2*epsilon*sqrt(pi*ds));
    else
        R = norm(r(m,:) - r(n,:));
        Zmn(m,n) = 1/(4*pi*epsilon*R);
    end

end

end

V = zeros(N,1);
V(1:M) = V0/2;
V(M+1:N) = -V0/2;

Q = Zmn\V;

rho_top = Q(1:M)/ds;

rho_top_grid = reshape(rho_top, [nx, ny])';

x = reshape(r(1:M,1), [nx, ny])';
y = reshape(r(1:M,2), [nx, ny])';

figure;
surf(x, y, rho_top_grid);

xlabel('x [m]');
ylabel('y [m]');
zlabel('\rho_s [C/m^2]');
title('Surface Charge Density on Top Plate');

shading interp;
colorbar;
grid on;

```

The code generates the following 3D representation:

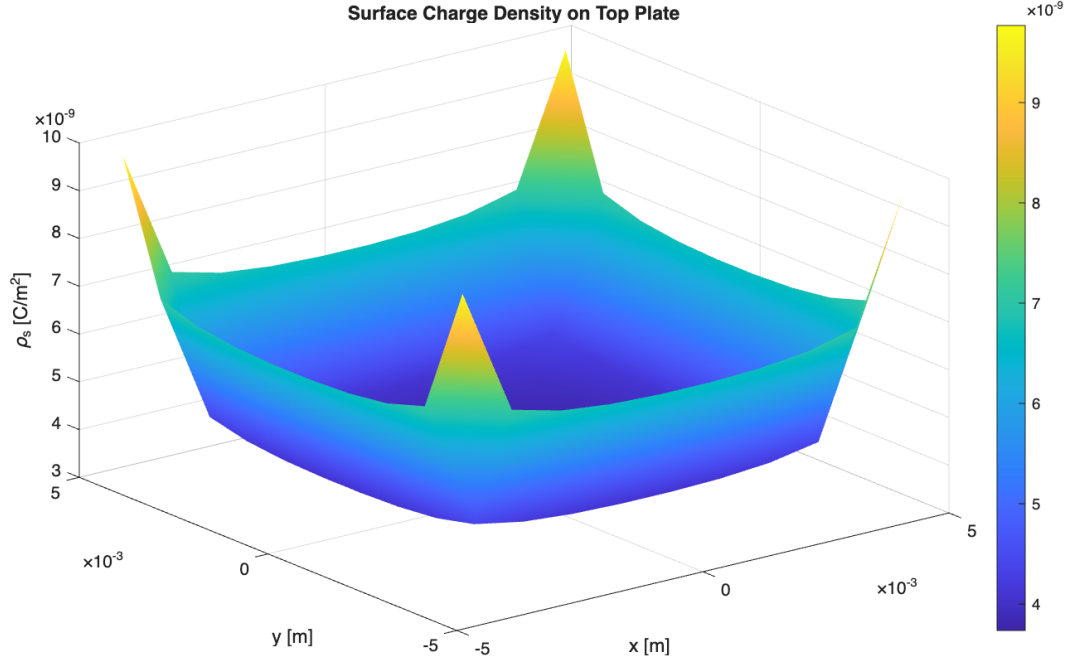


Figure 3: Surface Charge Density on Top Plate

We can experiment how different patch sizes affect the result by using $n_x, n_y = 5 \Rightarrow dx, dy = 2 \text{ mm}$ and $n_x, n_y = 20 \Rightarrow dx, dy = 0.5 \text{ mm}$ instead of our previous patch sizes of 10.

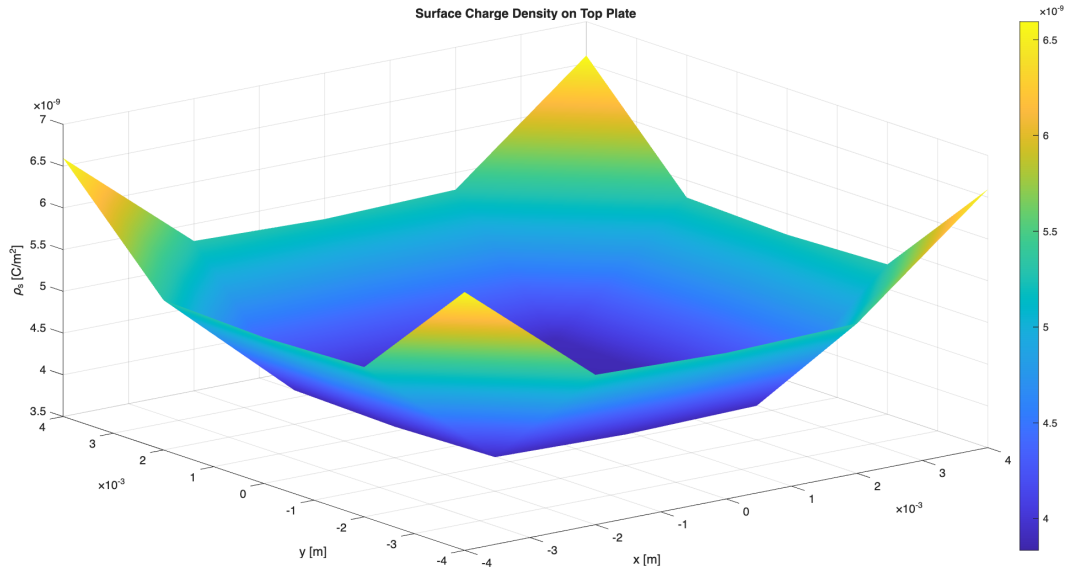


Figure 4: Surface Charge Density on Top Plate with Patch Size of 5

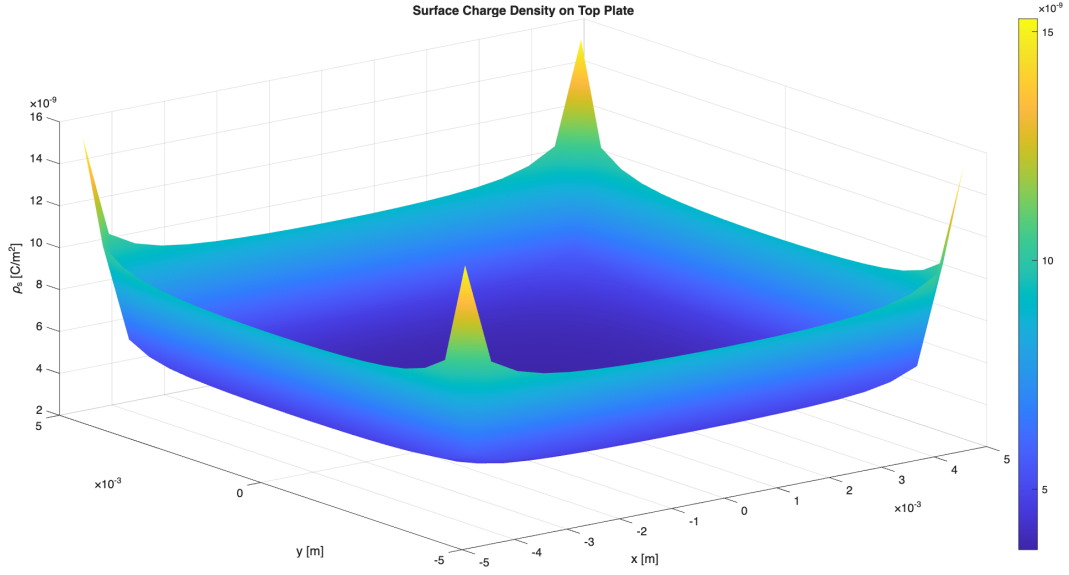


Figure 5: Surface Charge Density on Top Plate with Patch Size of 10

When we use a smaller patch size, we see that the figure becomes less continuous and more discretized, leading to a less smooth representation. On the other hand, making the patch size too large stretches the 3D representation and takes a little longer to compute. Thus, the optimal patch size for a good balance of computation time, smoothness, and sizing is $n_x, n_y = 10$, as used in the MATLAB code above.

4 Dependence of the Capacitance on Plate Separation

Finally, we investigate how the capacitance of the parallel plate capacitor varies depending on the distance between the two plates. We achieve this by assuming that the width $w = 1$ cm and the separation d varies between $0.1 \text{ mm} \leq d \leq 4 \text{ cm}$. We then plot every point in this dependence and compare it to the capacitance of a parallel plate capacitor obtained via the approximate formula $C = \frac{\epsilon_0 w^2}{d}$. We find each point in this dependence by solving the formulated integral equation for the discretized surface charges and then find their respective capacitances.

We use the following MATLAB code to find and plot the dependence:

```
clear; clc; close all;

epsilon = 8.85e-12;
V0 = 1;

w = 1e-2; % 1 cm
d_values = linspace(0.1e-3, 10e-3, 100);

nx = 20;
ny = 20;

dx = w/nx;
dy = w/ny;
ds = dx*dy;

M = nx*ny;
N = 2*M;

C_num = zeros(size(d_values));
C_approx = epsilon*w^2 ./ d_values;

for id = 1:length(d_values)

    d = d_values(id);

    r = zeros(N,3);
    cnt = 0;

    for k = 1:2
        for iy = 1:ny
            for ix = 1:nx
                cnt = cnt + 1;

                r(cnt,1) = -w/2 + (ix - 0.5)*dx;
                r(cnt,2) = -w/2 + (iy - 0.5)*dy;

                if k == 1
                    r(cnt,3) = d/2;
                else
                    r(cnt,3) = -d/2;
                end
            end
        end
    end
```

```

end

Zmn = zeros(N,N);

for m = 1:N
    for n = 1:N

        if m == n
            Zmn(m,n) = 1/(2*epsilon*sqrt(pi*ds));

        elseif abs(r(m,1)-r(n,1)) < 1e-15 && abs(r(m,2)-r(n,2)) < 1e-15
            Zmn(m,n) = (1/(2*epsilon*ds))*(-d + sqrt(ds/pi + d^2));

        else
            R = norm(r(m,:) - r(n,:));
            Zmn(m,n) = 1/(4*pi*epsilon*R);
        end

    end

end

V = zeros(N,1);
V(1:M) = V0/2;
V(M+1:N) = -V0/2;

Q = Zmn\V;

C_num(id) = sum(Q(1:M))/V0;

end

figure;
plot(d_values*1000, C_num, '- ', 'LineWidth', 1.5);
hold on;
plot(d_values*1000, C_approx, '—', 'LineWidth', 1.5);

xlim([0 10]);

xlabel('Plate separation d [mm]');
ylabel('Capacitance C [F]');
title('Capacitance vs Plate Separation');
legend('Numerical integral equation result', ...
        'Approximate C = \epsilon_0 w^2/d');
grid on;

```

The above code yields the graph:

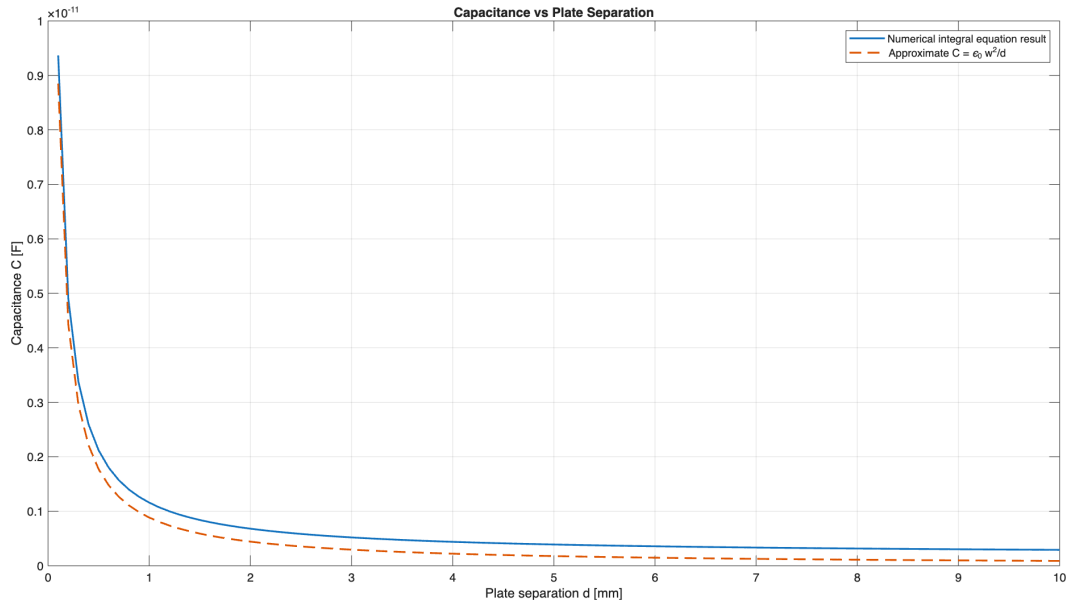


Figure 6: Plot of Capacitance vs. Plate Separation using the Integral Equation Result (solid line) and the Approximation Result (dashed line)

From this capacitance vs. plate separation plot, we can see the approximate formula $C = \frac{\epsilon_0 w^2}{d}$ is closer to the results obtained via the numerical solution when $d \ll w$. Visually, from this graph, we see that is when $d \lesssim 0.5$ mm.

5 Conclusion

By using MATLAB, we can calculate numerical answers to extract the capacitance of a parallel plate capacitor with given voltages. By formulating an integral equation, discretizing the charges to create a matrix equation, and then utilizing MATLAB, we found the discretization of charges, the capacitance, and created two figures to visualize the surface charge density and dependence on capacitance vs. plate separation.